

Qns	Answer	Marks
7(a)	Transition emits (one) photon with energy equal to the difference in energy between the two levels	B1
	photon energy = $h \times$ (frequency of radiation)	B1
	Thus, single frequency of radiation for each transition	
7(b)(i)	line to the left of the pair in Fig. 8.2, labelled A AND larger gap between line A and the nearest of the pair in Fig. 7.2, than between the lines in the pair	B1
7(b)(ii)	line to the left of both the pair in Fig. 7.2 and line A, labelled B AND larger gap between line B and line A, than between line A and the nearest one of the pair in Fig. 7.2	B1
7(c)	$\Delta E = E_{\text{photon}} = hf$ $(E_3 - E_1) = (E_2 - E_1) + (E_3 - E_2) = hf_A + hf_B$	C1
	$E_3 = E_1 + h(f_A + f_B)$	A1

